Q1
pttest OUTPUT-

```
$ pttest

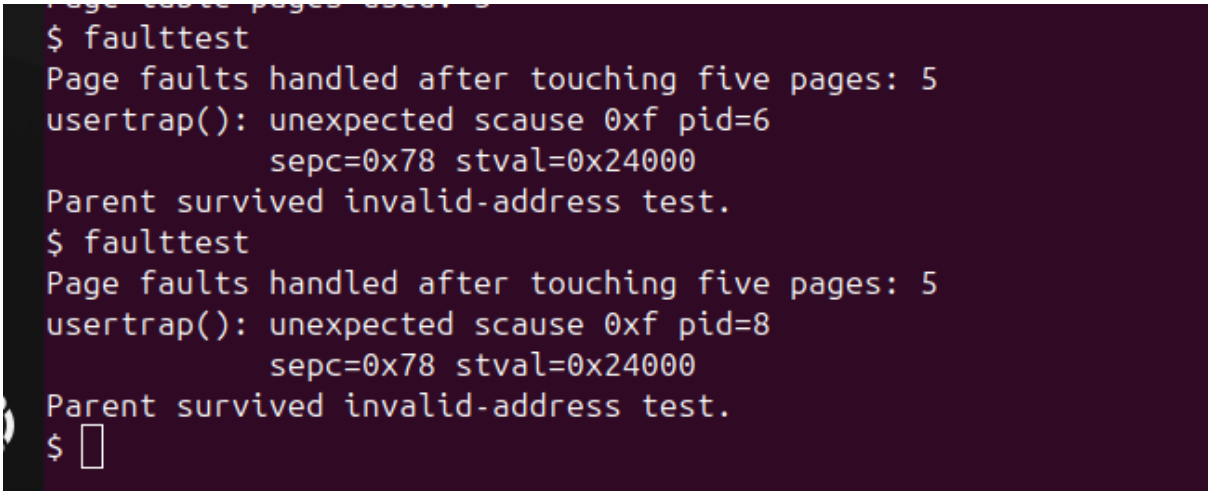
--- Page table before sbrk ---
Page table for pid 4:
Virtual address       Physical address      P W U
VA 0x0 PA 0x87f22000 P=1 W=0 U=1
VA 0x1000 PA 0x87f1f000 P=1 W=1 U=1
VA 0x3000 PA 0x87f1d000 P=1 W=1 U=1
Total valid user pages: 3
Page-table pages used: 3

--- Page table after sbrk(4 pages) ---
Page table for pid 4:
Virtual address       Physical address      P W U
VA 0x0 PA 0x87f22000 P=1 W=0 U=1
VA 0x1000 PA 0x87f1f000 P=1 W=1 U=1
VA 0x3000 PA 0x87f1d000 P=1 W=1 U=1
VA 0x4000 PA 0x87f25000 P=1 W=1 U=1
VA 0x5000 PA 0x87f53000 P=1 W=1 U=1
VA 0x6000 PA 0x87f49000 P=1 W=1 U=1
VA 0x7000 PA 0x87f46000 P=1 W=1 U=1
Total valid user pages: 7
Page-table pages used: 3

--- Page table after touching the four pages ---
Page table for pid 4:
Virtual address       Physical address      P W U
VA 0x0 PA 0x87f22000 P=1 W=0 U=1
VA 0x1000 PA 0x87f1f000 P=1 W=1 U=1
VA 0x3000 PA 0x87f1d000 P=1 W=1 U=1
VA 0x4000 PA 0x87f25000 P=1 W=1 U=1
VA 0x5000 PA 0x87f53000 P=1 W=1 U=1
VA 0x6000 PA 0x87f49000 P=1 W=1 U=1
VA 0x7000 PA 0x87f46000 P=1 W=1 U=1
Total valid user pages: 7
Page-table pages used: 3
$
```

Q2

faulttest OUTPUT-



Q3

pagerepl OUTPUT-

```
Parent survived invalid-address test.
$ pagerepl
Reference string: 7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1

FIFO trace (3 frames):
7: [ 7 - - ] FAULT
0: [ 7 0 - ] FAULT
1: [ 7 0 1 ] FAULT
2: [ 2 0 1 ] FAULT
0: [ 2 0 1 ] HIT
3: [ 2 3 1 ] FAULT
0: [ 2 3 0 ] FAULT
4: [ 4 3 0 ] FAULT
2: [ 4 2 0 ] FAULT
3: [ 4 2 3 ] FAULT
0: [ 0 2 3 ] FAULT
3: [ 0 2 3 ] HIT
2: [ 0 2 3 ] HIT
1: [ 0 1 3 ] FAULT
2: [ 0 1 2 ] FAULT
0: [ 0 1 2 ] HIT
1: [ 0 1 2 ] HIT
7: [ 7 1 2 ] FAULT
0: [ 7 0 2 ] FAULT
1: [ 7 0 1 ] FAULT

LRU trace (3 frames):
7: [ 7 - - ] FAULT
0: [ 7 0 - ] FAULT
1: [ 7 0 1 ] FAULT
2: [ 2 0 1 ] FAULT
0: [ 2 0 1 ] HIT
3: [ 2 0 3 ] FAULT
0: [ 2 0 3 ] HIT
4: [ 4 0 3 ] FAULT
2: [ 4 0 2 ] FAULT
3: [ 4 3 2 ] FAULT
0: [ 0 3 2 ] FAULT
3: [ 0 3 2 ] HIT
2: [ 0 3 2 ] HIT
1: [ 1 3 2 ] FAULT
2: [ 1 3 2 ] HIT
0: [ 1 0 2 ] FAULT
1: [ 1 0 2 ] HIT
7: [ 1 0 7 ] FAULT
0: [ 1 0 7 ] HIT
1: [ 1 0 7 ] HIT

Results for 3 frames (20 references):
FIFO faults=15 hits=5 hit ratio=5/20
LRU  faults=12 hits=8 hit ratio=8/20

Frame-count comparison:
Frames  FIFO faults/hit%       LRU faults/hit%
3       15/25%              12/40%
4       10/50%              8/60%
5       9/55%              7/65%
$
```